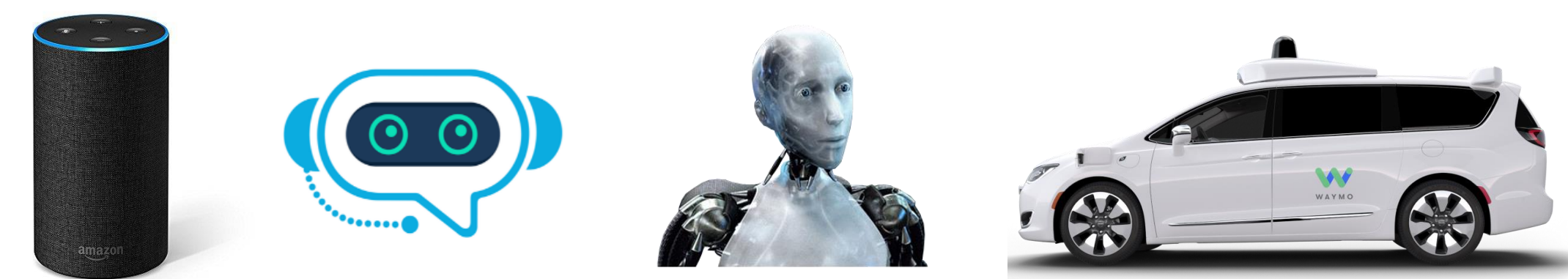


How would a non-expert assess the limits and capabilities of an AI system?

INTRODUCTION

Objective: Learn an interpretable model of a black-box agent by interrogating it.



Key technical challenge:

- Which sequence of queries to ask?

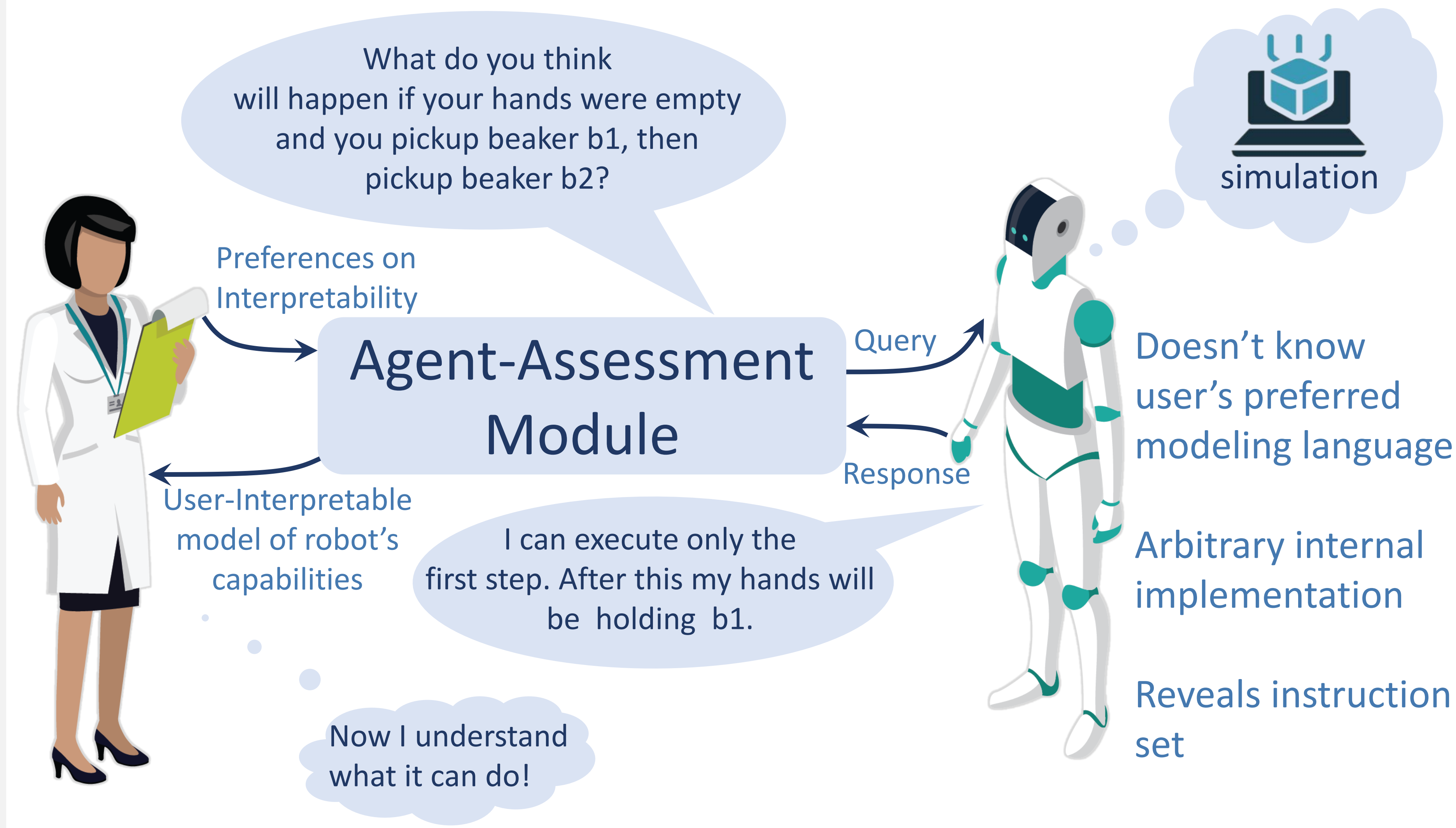
ALGORITHM (AIA)

- 1 Start with the most abstracted node in lattice.
- 2 Pick abstraction candidates in some order.
- 3 For each candidate, generate three models and for each pair of models:
- 4 • Generate a distinguishing query Q and pose it to the agent.
- 5 • Get the response R from the agent.
- 6 • Prune out the incorrect variants of candidate models.
- 7 • Repeat steps 3-6 till the model is fully estimated.
- 8 Return the final set of model(s).

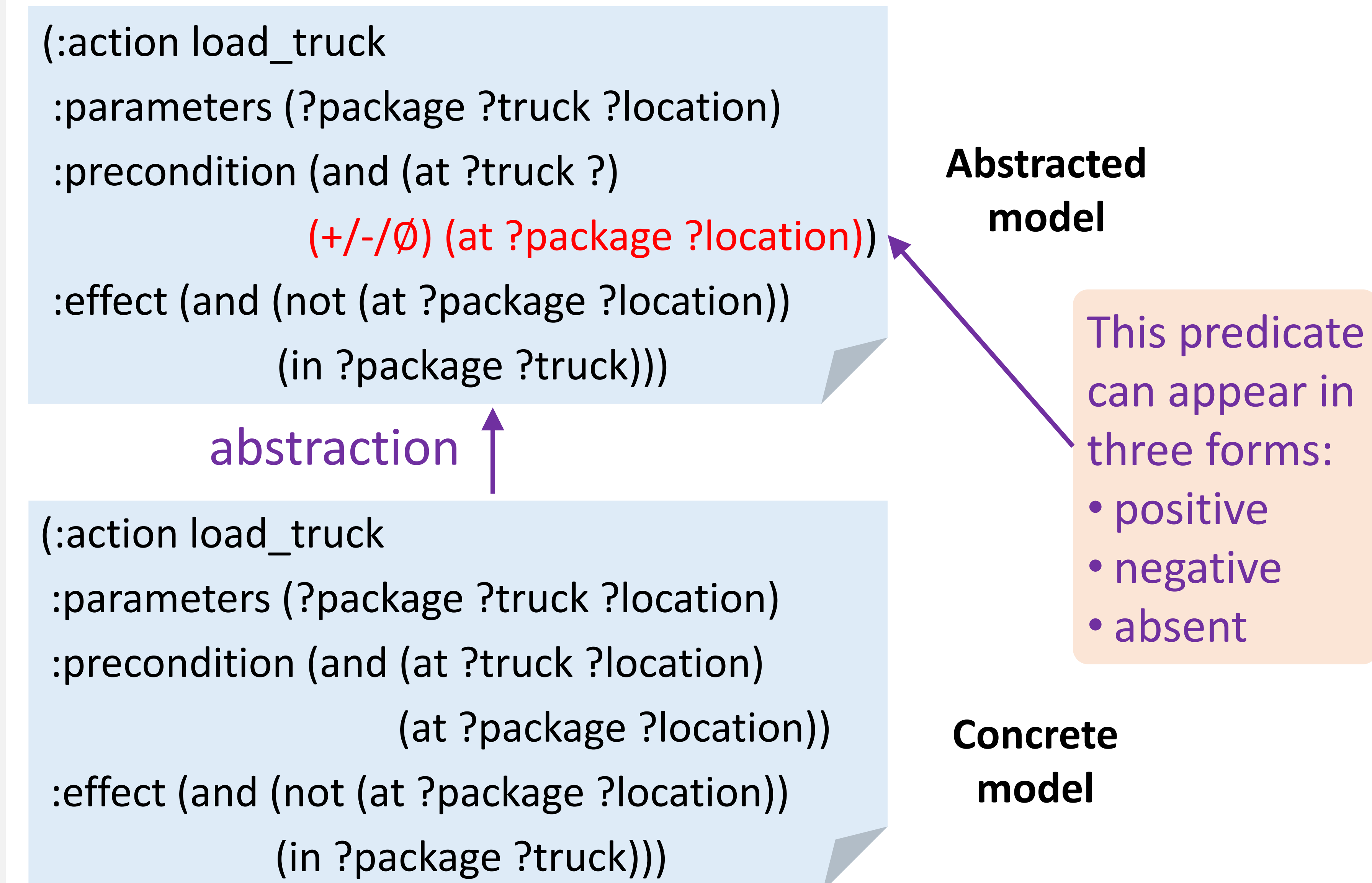
SALIENT FEATURES

- Efficiently learns internal model of an autonomous agent in a STRIPS-like form.
- Needs no prior knowledge of the agent model.
- Only requires an agent to have rudimentary query answering capabilities.
- Queries can be answered using a simulator.

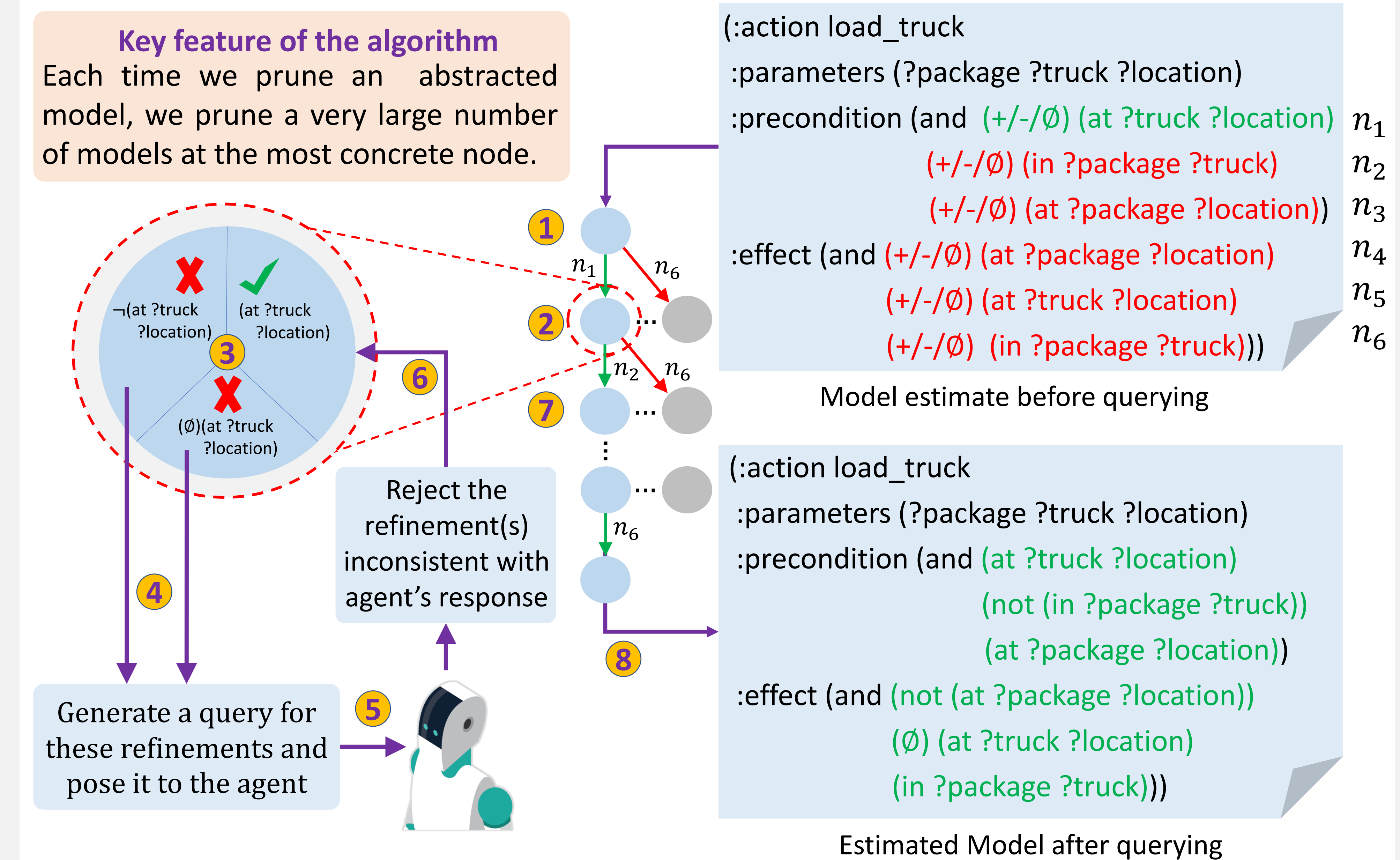
EXAMPLE OF AGENT INTERROGATION



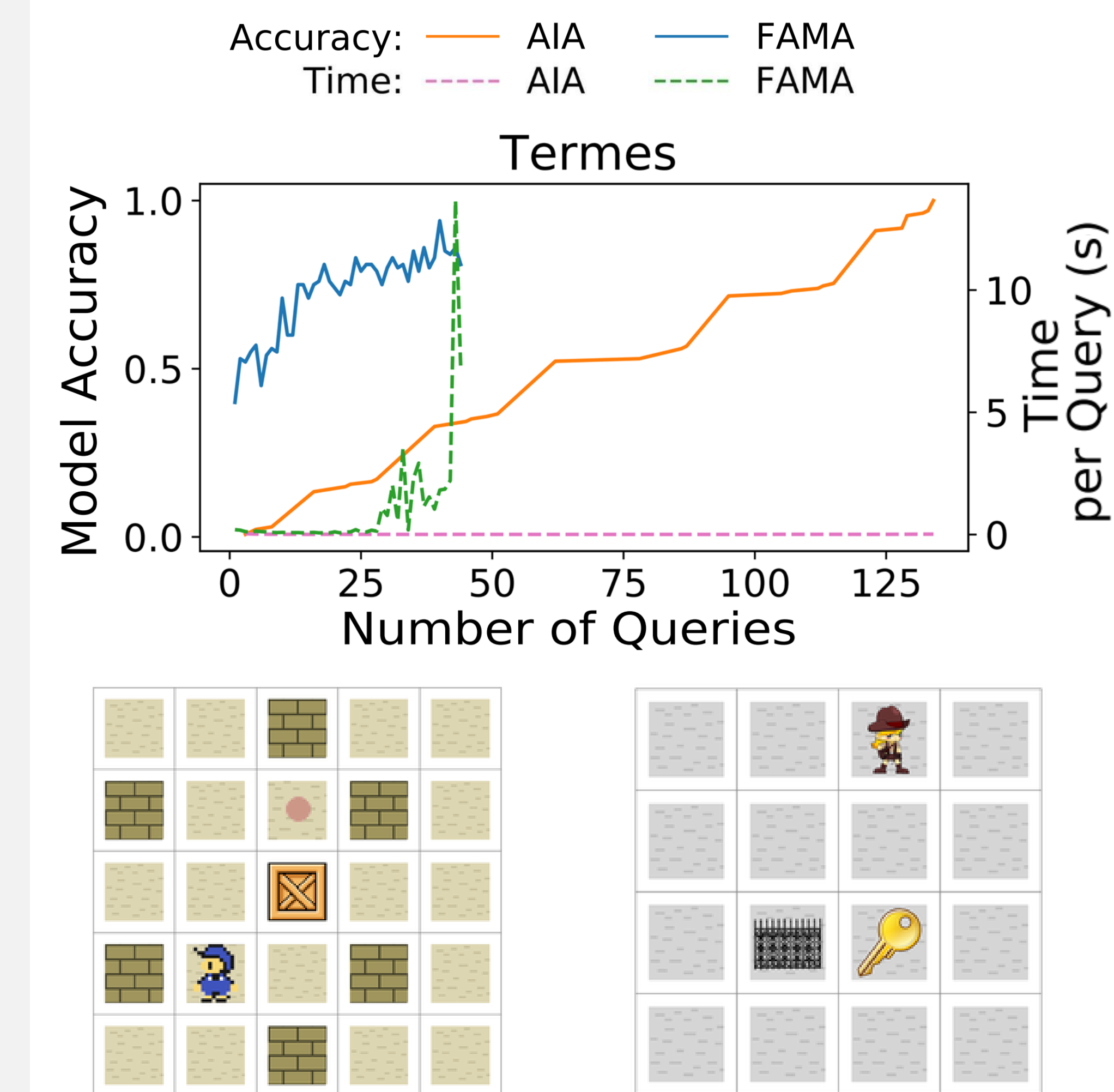
ABSTRACTION IN SPACE OF MODELS



KEY ALGORITHMIC PRINCIPLE



RESULTS



- AIA efficiently derives interpretable agent models for a range of agents.
- AIA is much faster than state of the art methods for deriving models based on passive observations.
- AIA offers better convergence guarantees.

Refer to the paper
for detailed results

bit.ly/3p4cVRu

